(a) Language → structured spatial goals

帮我看看二楼厨房台面附近的微波炉、二楼客厅沙发旁的茶几和
三楼客厅的电视是不是都还在原位，再顺路确认一下二楼卫生间马桶旁还有没有厕纸，先后顺序你自己安排。

microwave
L2 / kitchen
near(countertop)

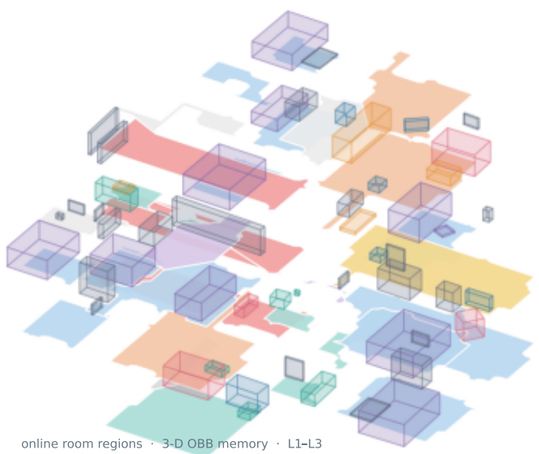
coffee table
L2 / living
near(sofa)

television
L3 / living
instance

toilet paper
L2 / bathroom
near(toilet)

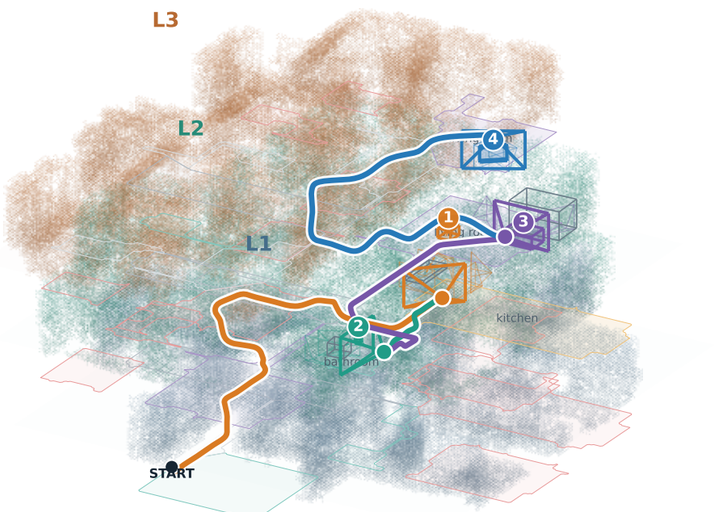
Qwen · 4 independent goals · no prescribed order · geometry chooses execution sequence

(b) Pre-explored 3-D semantic memory



online room regions · 3-D OBB memory · L1–L3

(c) Scene-centric task–viewpoint–path reasoning

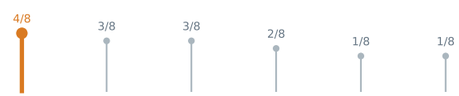


faint frustums: feasible q · saturated frustums: selected q* · colored curves: executed B-spline

(d) Spatial grounding and executable views



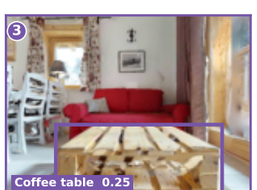
near(countertop) → surrounding region → q = (x, y, z, ψ)



(f) Open-vocabulary verification · 4 / 4



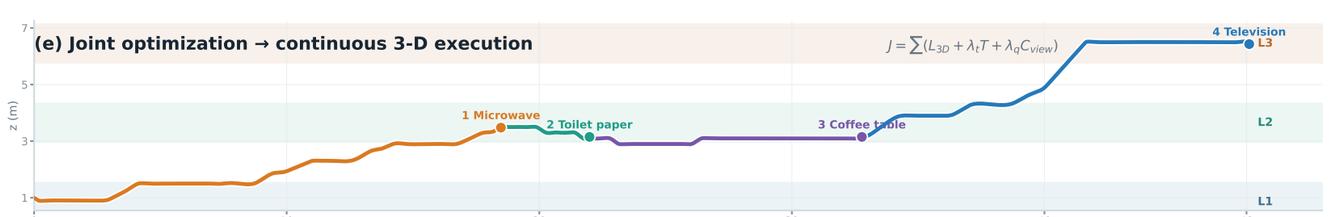






(e) Joint optimization → continuous 3-D execution

$$J = \sum (L_{3D} + \lambda_t T + \lambda_q C_{view})$$



16 feasible views | 3-D A* + B-spline | path 48.1 m | min. predicted clearance 0.10 m | 13.0 s